

INTERNSHIP REPORT

Of

Cisco Virtual Internship Program

Submitted to

School of Computer Science and Engineering

IILM University, Greater Noida, Uttar Pradesh

In partial fulfilment of the requirements for the degree of

**BACHELOR OF TECHNOLOGY IN COMPUTER SCIENCE AND
ENGINEERING**



Submitted by

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Roll Number: 2410031648

Batch: 2024 - 2028

Internship Organisation: Cisco Virtual Internship Program

Internship Duration: June 2026 to August 2026

Faculty Mentor: Dr. Swati Vashisht

Academic Session 2026-27

CANDIDATE'S DECLARATION

I, Priyanshu Yadav, Roll Number 2410031648, hereby declare that this internship report, titled “NetSage AI: Intelligent Network Troubleshooting Assistant”, is an original account of the work completed by me during the Cisco Virtual Internship Program from June 2026 to August 2026. It is submitted in partial fulfilment of the requirements for the degree of Bachelor of Technology in Computer Science and Engineering at IILM University, Greater Noida.

The implementation, analysis, and conclusions in this report are based on my project work. The documentation and learning resources consulted during development have been acknowledged in the references section. I understand that I am responsible for the accuracy of this submission and for compliance with academic-integrity requirements.

Signature: _____

Student Name: Priyanshu Yadav

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ACKNOWLEDGEMENT

I sincerely thank IILM University, Greater Noida, and the School of Computer Science and Engineering for giving me the opportunity to undertake this internship project. I am especially grateful to my faculty mentor, Dr. Swati Vashisht, for her guidance and encouragement during the preparation of this project and report.

I acknowledge the Cisco Virtual Internship Program for providing a practical learning context in networking. The internship helped me connect classroom concepts with the design of a software-based troubleshooting assistant.

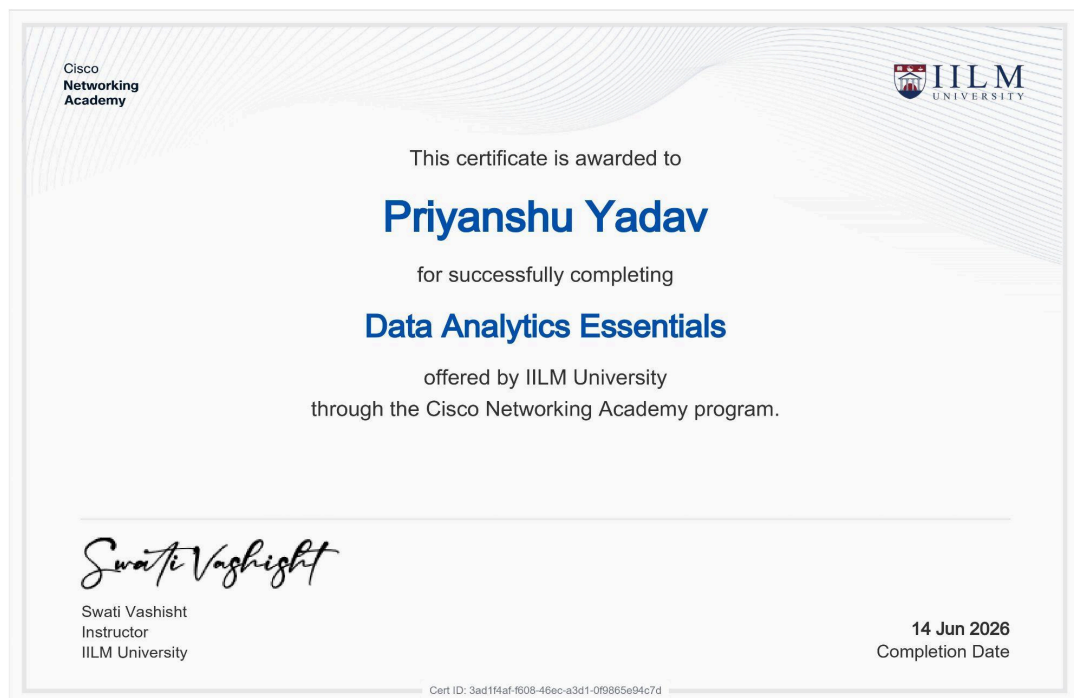
I also thank my family and peers for their support throughout the internship period. Their encouragement helped me remain focused while designing, testing, deploying, and documenting NetSage AI.

Signature: _____

Student Name: Priyanshu Yadav

Roll Number: 2410031648

CISCO COURSE COMPLETION CERTIFICATE



This certificate is awarded to

Priyanshu Yadav

for successfully completing

Introduction to Modern AI

offered by IILM University
through the Cisco Networking Academy program.

Swati Vashisht

Swati Vashisht
Instructor
IILM University

14 Jun 2026
Completion Date

Cert ID: 3a33ad86-ffec-47bc-a526-fdac4044e44f

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4. PROJECT DESCRIPTION

4.1 Introduction

Computer networks support communication, cloud access, business applications, and secure information exchange. When connectivity fails, a network engineer must examine symptoms, device state, addressing details, and command output. This can be time-consuming because a single symptom may be caused by several different problems. A failed connection to a server, for example, may result from an interface issue, an incorrect IP configuration, a missing VLAN, a routing error, or an access-control policy.

NetSage AI was developed as a practical troubleshooting assistant for common network faults. The application accepts a structured case containing symptoms, topology notes, and evidence such as show-command output. It performs deterministic checks and presents a structured diagnosis containing a probable root cause, evidence, a recommended fix, and a verification method.

The project treats AI as decision support rather than an autonomous operator. A human reviewer can accept, edit, or reject every diagnosis. This preserves accountability and makes the tool suitable for learning-oriented use.

4.2 Organisation Profile

This project was completed in the context of the Cisco Virtual Internship Program. The internship provided a setting for applying networking concepts through an applied software project. The project focused on a structured approach to network diagnosis rather than on direct changes to live production devices.

The work connects common networking topics with an accessible software workflow. It demonstrates how evidence, rules, and AI-assisted explanation can be combined to support a junior engineer or student during fault analysis.

4.3 Problem Statement

Network troubleshooting is frequently manual. Engineers inspect configurations, compare command output, and use experience to identify probable causes. This is difficult for junior engineers because similar symptoms can arise from faults at

different layers. An incomplete investigation can cause delay, unnecessary escalation, or an unsafe fix.

The problem addressed by NetSage AI is the absence of a simple, explainable assistant for routine network-fault diagnosis. The system must check facts consistently, explain why it reached a conclusion, recommend a safe next action, and leave the final decision to a human reviewer. The first version covers duplicate IP addresses, incorrect masks, gateway mismatch, interface status, VLANs, routes, DHCP, DNS, ACLs, NAT, and wireless conditions.

4.4 Project Objectives

- Create a reusable case library containing more than 30 troubleshooting cases with symptoms, evidence, expected fault labels, and structured payloads.
- Implement deterministic checks for frequent Layer 2, Layer 3, and service-level faults.
- Generate a clear diagnosis with root cause, supporting evidence, recommended action, confidence level, and verification steps.
- Provide a Streamlit dashboard for case selection, result review, and human feedback.
- Demonstrate responsible AI through Accept, Edit, and Reject review actions.

4.5 Scope of the Project

NetSage AI is an educational and internship-ready prototype. It works with curated troubleshooting cases instead of connecting to production devices. Every case links readable evidence with a JSON payload so that the rule engine can operate on consistent input. The current library includes VLAN, addressing, routing, DHCP, DNS, ACL, NAT, and wireless scenarios.

The application does not log in to network equipment, apply configuration changes, collect credentials, or replace the judgment of a network engineer. Live AI diagnosis is optional. When an API key is unavailable, the application continues with a local structured fallback so that the main workflow remains demonstrable.

4.6 Technologies and Tools Used

Technology / Tool	Purpose	Reason for Selection
Python	Core application and rule checks	Readable and suitable for data processing.
Streamlit	Dashboard and review interface	Creates an interactive app directly from Python.
Pandas with CSV/JSON	Case library and evidence payloads	Keeps data simple to inspect and extend.
OpenAI Responses API	Optional structured AI diagnosis	Supports schema-based outputs when configured.
unittest	Automated checks	Verifies rule-engine and AI-path behavior.
GitHub and Streamlit Community Cloud	Source hosting and deployment	Provides a simple shareable deployment path.

Table 4.1: Technologies and tools used in NetSage AI

4.7 System Architecture

The architecture separates deterministic network evidence from AI-generated explanation. This division improves traceability because the diagnosis is grounded in findings produced by explicit checks.

Stage	Input / Output	Responsibility
1. Case selection	Symptoms, topology notes, show output, JSON payload	Loads a troubleshooting scenario.
2. Rule checker	Payload to structured findings	Identifies known faults and evidence.
3. Diagnosis engine	Prompt and findings to diagnosis	Creates root-cause explanation and verification plan.
4. Streamlit dashboard	Findings and diagnosis to interface	Displays charts, evidence, and review controls.
5. Human review log	Accept, Edit, or Reject to record	Keep the final decision with the reviewer.

Table 4.2: NetSage AI system architecture

Case data → Deterministic rule checker → Structured diagnosis → Human review → Responsible-AI log

Figure 4.1: High-level processing flow of NetSage AI

4.8 Methodology

An iterative development approach was followed. Each stage produced a testable output before the next stage was added. This ensured that the system had an evidence layer before AI-assisted diagnosis was introduced.

Phase	Activity	Output	Validation
Requirement analysis	Mapped requirements to data, rules, dashboard, diagnosis, and review.	Project scope and folder structure.	Deliverables matched to requirements.
Dataset design	Prepared 32 cases with CSV records and JSON payloads.	Case library with expected faults.	Coverage checked across concepts.
Rule implementation	Implemented checks for addressing, interfaces, VLANs, routes, and services.	Evidence-based findings.	Unit tests checked expected results.
AI integration	Prepared prompt and optional structured diagnosis path.	Root-cause and remediation draft.	Fallback retained for missing API configuration.
Dashboard and review	Built selection, findings, charts, and review actions.	Interactive Streamlit app.	Representative cases tested manually.
Deployment	Published the prototype through Streamlit Community Cloud.	Shareable web application.	App launched and checked online.

Table 4.3: Development methodology followed for the project

The deterministic rule checker and AI integration were verified through automated tests. During development, 23 automated tests passed. The dashboard was manually checked with cases drawn from different categories, including VLAN, routing, DHCP, DNS, and wireless troubleshooting.

4.9 Expected Outcomes

The expected outcome is a repeatable troubleshooting workflow for students and junior engineers. Rather than starting with an unstructured guess, the user can examine the symptom, inspect rule findings, read a concise diagnosis, and verify the recommended correction. The case library also supports revision because every scenario connects a fault with evidence.

- Faster identification of common faults in curated network scenarios.
- More consistent troubleshooting through explicit evidence and rules.
- Improved learning through explanations, recommended fixes, and verification steps.
- Human oversight through documented Accept, Edit, and Reject decisions.
- A deployable prototype that can be demonstrated using a live Streamlit link.

Future work can add persistent review storage, direct laboratory integration, and a formal AI-human agreement metric. The prototype should be evaluated by diagnosis correctness for known cases, the usefulness of verification steps, and the reviewer's ability to challenge or revise the result.

4.10 Certificates of Completion and Other Communication Proof

Cisco Networking Academy <noreply@netacad.com>
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Sun, Jun 14, 2026 at 4:21 PM

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that this is just the beginning of your successful journey.

Proudly applauding your hard work,
The Cisco Networking Academy Team

[Quoted text hidden]



Well Done, Priyanshu Yadav! Your Certificate of Completion is Ready

3 messages

Cisco Networking Academy <noreply@netacad.com>
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